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Prof. M. G. D. Geers  
Editor-in-Chief  
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Dear Editor,

I am pleased to submit my manuscript entitled “**Granular penetration of a growing morphoelastic root under variable gravitational acceleration**” for consideration for publication in *European Journal of Mechanics – A/Solids*.

The manuscript studies how deep a growing root can penetrate a granular regolith when the gravitational acceleration varies, which is the regime that matters for cultivation in lunar, martian and other low-(super-)gravity environments and for compaction-driven rooting limits in agricultural soils. A growing root is modeled as a single morphoelastic rod with three active controls (gravitropism, circumnutation, intrinsic-curvature relaxation) that advances by a subapically localized growth law into a Mohr–Coulomb granular medium whose resistance is described by a classical Bekker-type cone-bearing capacity, with a finite generative-thrust ceiling that caps axial burial. The equilibrium is solved as a genuine two-point boundary-value problem by shooting, so the numbers in the paper are exact evaluations of the stated equations, not approximate fits.

The central result is a closed-form penetration law. Gravity enters the model in three separable places—the self-weight of the organ, the overburden gradient that sets the granular strength, and the gain of the gravitropic drive (a statolith-sedimentation picture)—and from their interplay an exact inverse law follows: the quasi-static burial depth scales as  $z^* \propto 1/g$ , with the coefficient fixed by cohesion, friction angle, bulk density, tip radius and generative thrust, and with an explicit threshold below which the medium is self-supporting and the root becomes a surface runner. The law is confirmed numerically over a 30-fold range of  $g$  spanning Europa to  $4g$ ; at low gravity penetration is limited instead by the grown length and by lateral wandering, so two clean regimes ( $z^* \propto 1/g$  where thrust-limited,  $z \sim L$  where length-limited) are isolated. A verification study separates the two control mechanisms and shows that gravitropism and circumnutation act as independent depth and shape controls, recovering straight, waving and coiling tip paths within one framework.

Works on root pattern formation on rigid, inclined planes (e.g. Sipos & Várkonyi, *JMPS* 160:104789, 2022; Porat *et al.*, *PNAS* 121:e2312761121, 2024) are closely related and are cited as such; they consider an organ sliding over an impenetrable interface, whereas the present work concerns burial into a yielding granular volume under variable gravity. The manuscript is written entirely on its own terms, with the equations, parameters and dimensionless scalings reported in full.

The model is deliberately minimal and cheap (a full gravity sweep runs in minutes on a single core) and is directly reusable for forward estimates of inhospitable-regolith rooting, making it, I hope, of interest to the mechanics community the journal serves. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no conflicts of interest.

Thank you for your time and consideration.

Sincerely,

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